Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Identify Z in the sequence:

$$\mathrm{CH_3CH_2CH=CH_2} \xrightarrow{\mathrm{HBr/H_2O_2}} \mathrm{Y} \xrightarrow{\mathrm{C_2H_5ONa}} \mathrm{Z}$$

(A) $\mathrm{(CH_3)_2\,COHCH_2CH_3}$

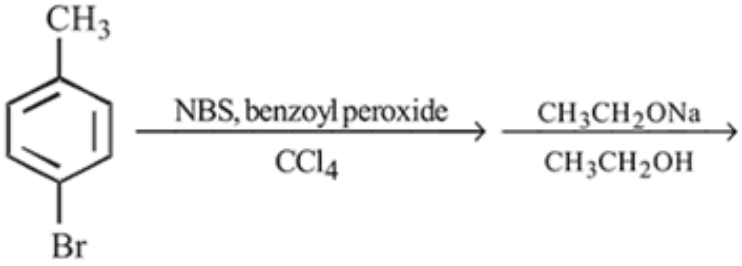
(B) $\mathrm{CH_3CH_2CH\,(CH_3)-O-CH_2CH_3}$

(C) $\mathrm{CH_3\,(CH_2)_3-O-CH_2CH_3}$

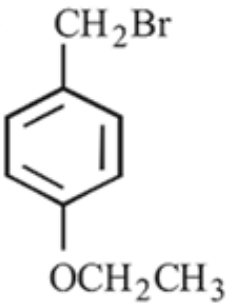
(D) $\mathrm{CH_3\,(CH_2)_4-O-CH_3}$

Q.2

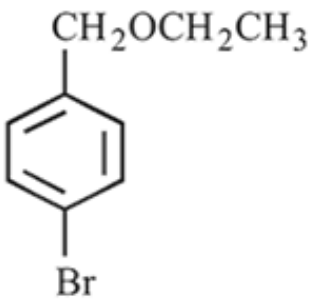
What is the major product of the following reaction?



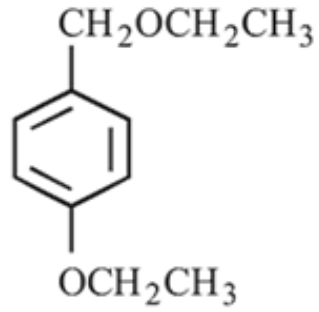
(A)

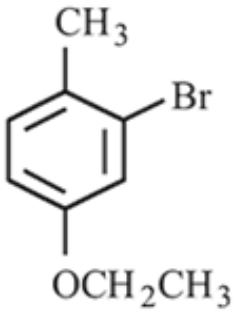


(B)

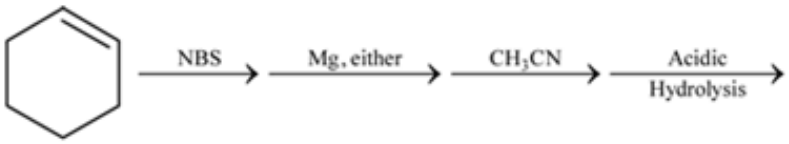


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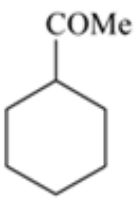
(C) 

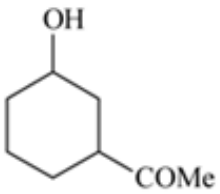
(D) 

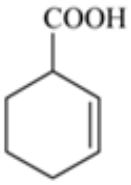
Q.3

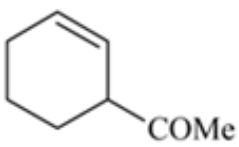


final product is:

(A) 

(B) 

(C) 

(D) 

Q.4

Which of the following cannot be made by using Williamson's synthesis?

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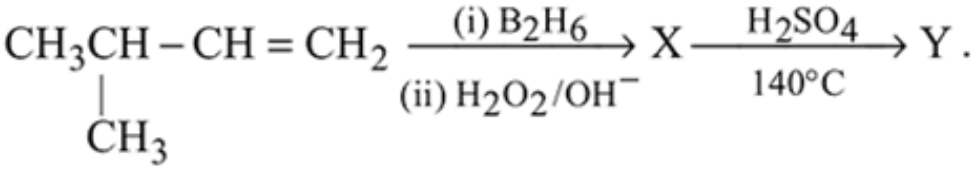
(A) Methoxybenzene

(B) Benzyl p -nitrophenyl ether

(C) Methyl tertiary butyl ether

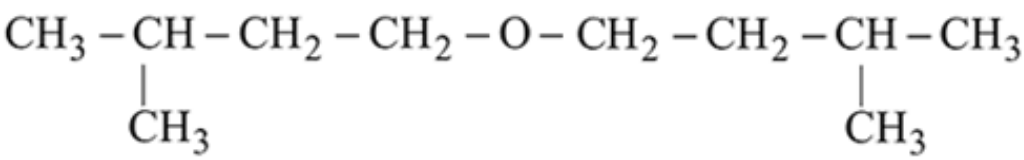
(D) Di-tert-butyl ether

Q.5

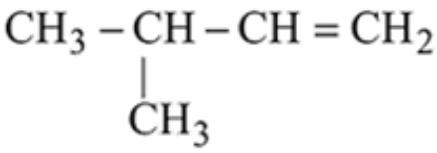


What is Y?

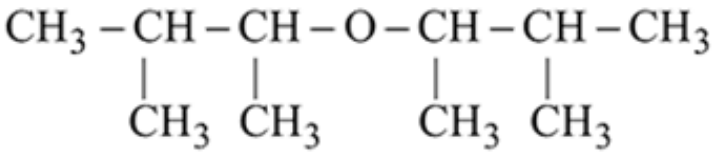
(A)



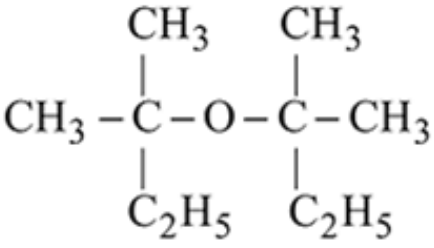
(B)



(C)



(D)



Q.6

Williamson's synthesis is used to prepare

(A) acetone

(B) diethyl ether

(C) P.V.C.

(D) bakelite

Q.7

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Williamson synthesis of ether is an example of:

(A) Nucleophilic addition

(B) Electrophilic addition

(C) Electrophilic substitution

(D) Nucleophilic substitution

Q.8

An ether (A), $C_5H_{12}O$, when heated with excess of hot concentrated HI produced two alkyl halides which when treated with $NaOH$ yielded compounds (B) and (C). Oxidation of (B) and (C) gave a propanone and an ethanoic acid respectively. The IUPAC name of the ether (A) is:

(A) 2 -ethoxypropane

(B) ethoxypropane

(C) methoxybutane

(D) 2-methoxybutane

Q.9

Which are not cleaved by HIO_4?
I. glycerol
II. glycol
III. 1,3 propenediol
IV. methoxy-2-propanol

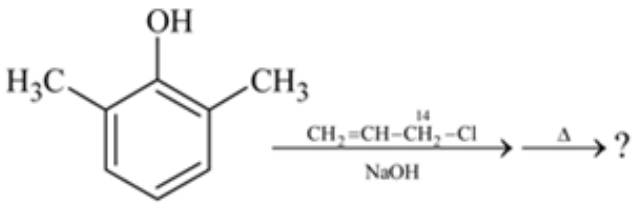
(A) I, II, III, IV

(B) I, II

(C) II, III

(D) III, IV

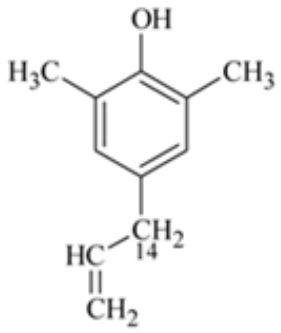
Q.10



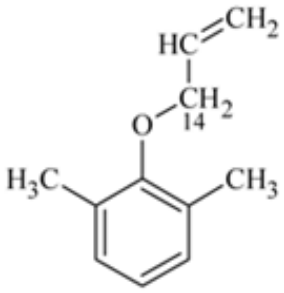
What is the missing product?

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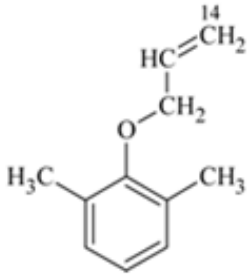
(A)



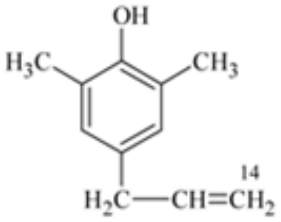
(B)



(C)



(D)



Q.11

An aromatic ether is not cleaved by HI even at $525\ K$. The compound is

(A) $C_6H_5OCH_3$

(B) $C_6H_5OC_6H_5$

(C) $C_6H_5OC_3H_7$

(D) Tetrahydrofuran

Q.12

Acetic anhydride reacts with diethyl ether in the presence of anhydrous $AlCl_3$ to give

(A) CH_3COOCH_3

(B) $CH_3CH_2COOCH_3$

(C) $CH_3COOCH_2CH_3$

(D) CH_3CH_2OH

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Q.13

In the following reaction.

$C_2H_5OC_2H_5 + 4H \xrightarrow{\text{Red P+HI}} 2X + H_2O;$

X is

(A) ethane

(B) ethylene

(C) butane

(D) propane

Q.14

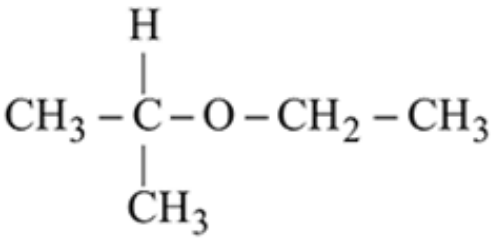
(X) $\xrightarrow{H_3O^+}$ Y + Z (Y and Z both give he Iodoform test).

$(C_5H_{10}O)$

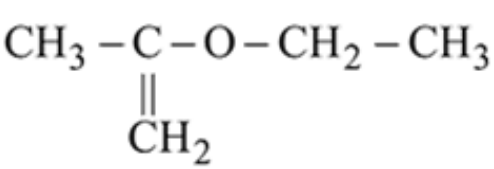
The compound X is $-$

(A) $CH_3 - CH = CH - O - CH_2 - CH_3$

(B)



(C)

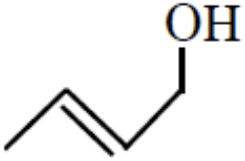


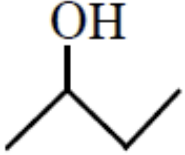
(D) Both (a) and (c)

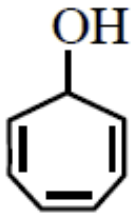
Q.15

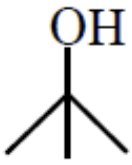
How many alcohols give immediate turbidity on heating with luca's Reagent?

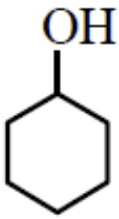
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(A) 

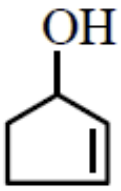
(B) 

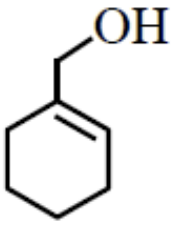
(C) 

(D) 

(E) 

(F) 

(G) 

(H) 

(A) 7

(B) 5

(C) 11

(D) 13

Answers & Solutions

Q.1 Answer:

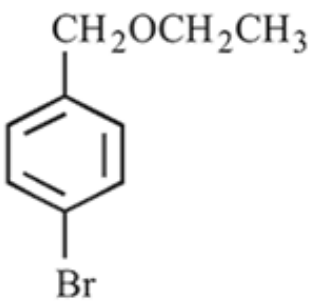
$CH_3\,(CH_2)_3 - O - CH_2CH_3$

Solution:

$$CH_3CH_2CH = CH_2 \xrightarrow{HBr/H_2O_2} CH_3CH_2CH_2CH_2Br \xrightarrow{C_2H_5ONa} CH_3\,(CH_2)_3\,OC_2H_5$$

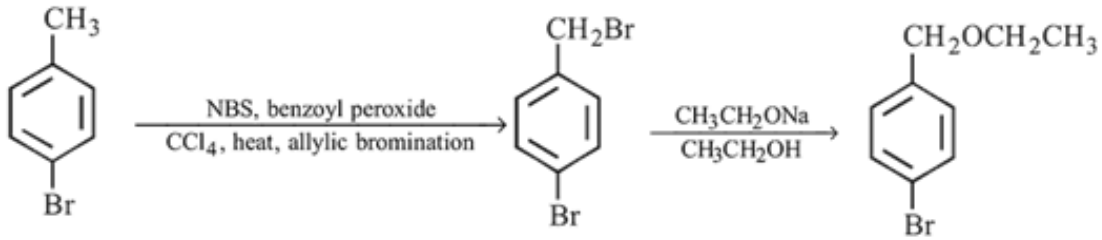
In presence of H_2O_2, HBr adds in anti-Markovnikov's way (peroxide effect).

Q.2 Answer:

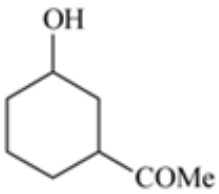


Solution:

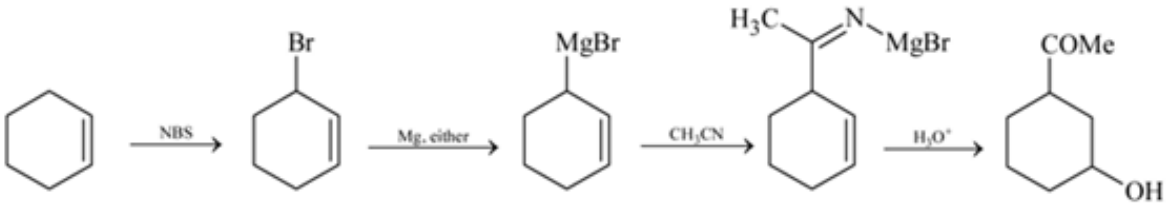
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Q.3 Answer:



Solution:



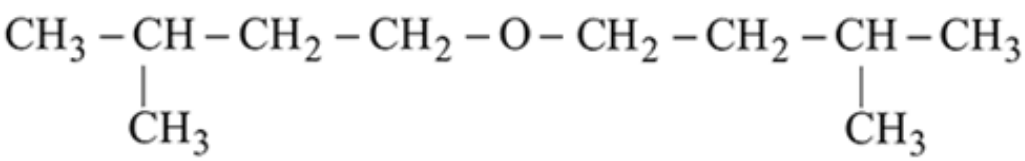
Q.4 Answer:

Di-tert-butyl ether

Solution:

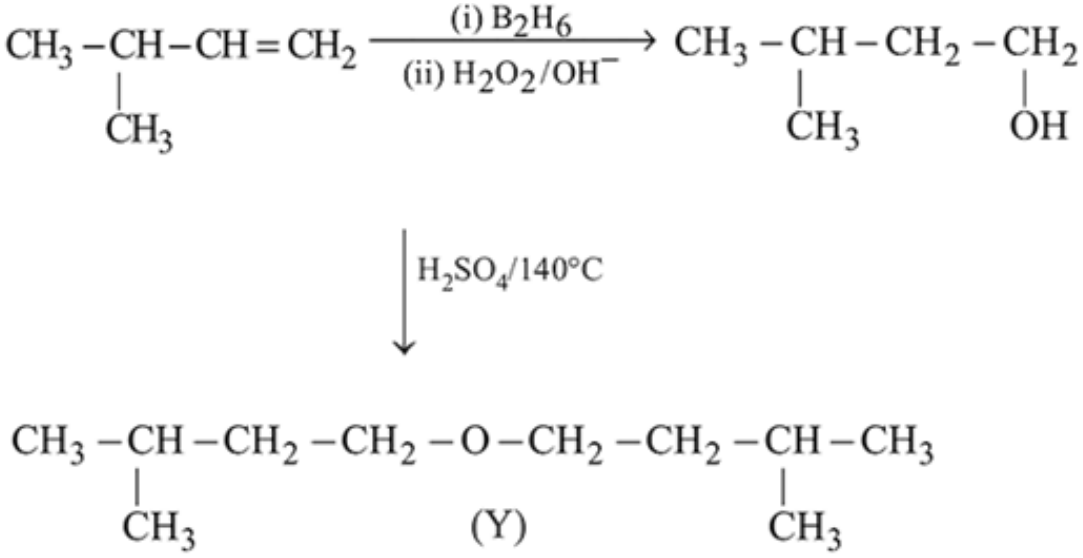
The two components should be $(CH_3)_3\,CONa$ $(CH_3)_3\,CBr$. However, tert-alkyl halides tend to undergo elimination reaction rather than substitution leading to the formation of an alkene, $Me_2C = CH_2$

Q.5 Answer:



Solution:

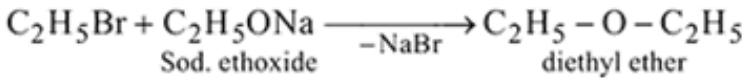
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Q.6 Answer:

diethyl ether

Solution:



Q.7 Answer:

Nucleophilic substitution

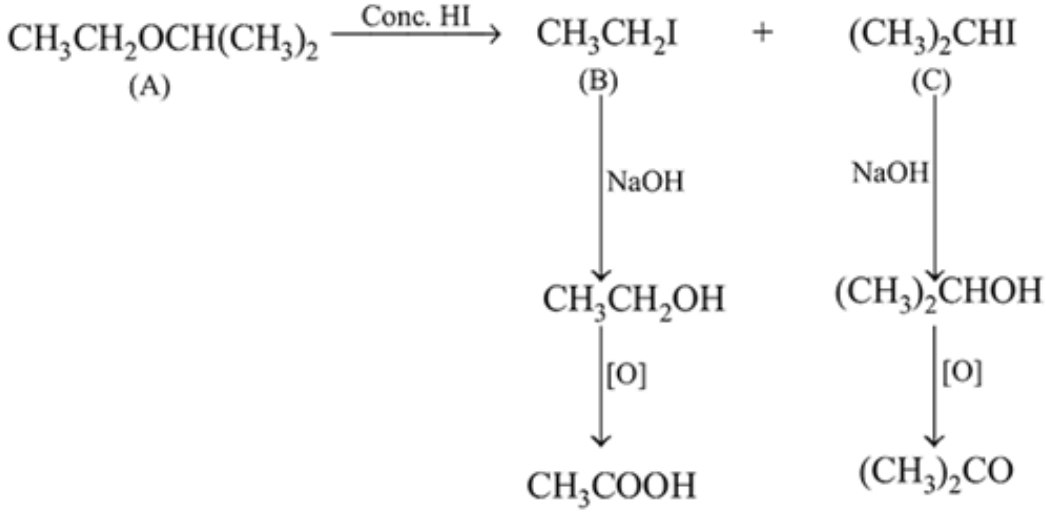
Solution:

This method is suitable for the preparation of a wide variety of unsymmetrical ethers. The nucleophilic substitution of halides with alkoxide leads to desired product.

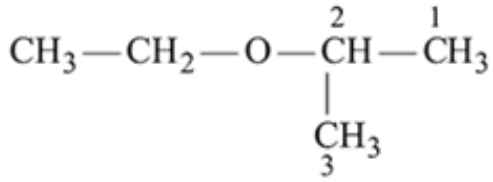
Q.8 Answer:

2 -ethoxypropane

Solution:



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hence the IUPAC name of compound is 

2–ethoxy propane

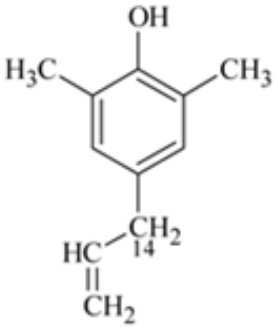
Q.9 Answer:

III, IV

Solution:

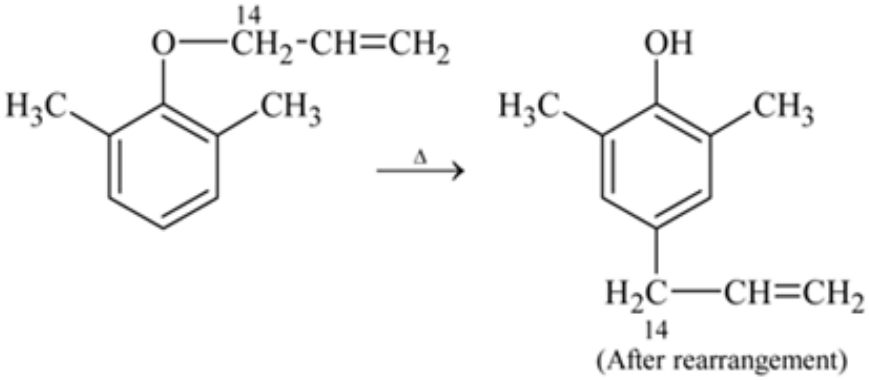
HIO_4 **will not oxidise diol from 1,3 atom and cannot used for cleavage of ether.**

Q.10 Answer:



Solution:

Claises Rearrangement



(After rearrangement)

Q.11 Answer:

$C_6H_5OC_6H_5$

Solution:

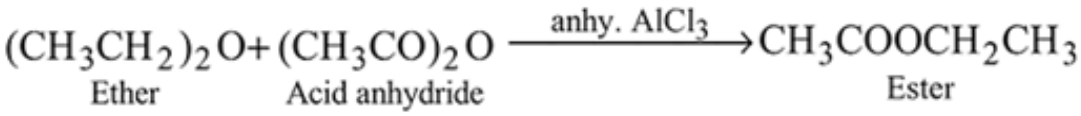
Due to greater electronegativity of sp^2 -hybridized carbon atoms of the benzene ring, diaryl ethers are not attacked by nucleophiles like I^-.

Q.12 Answer:

$CH_3COOCH_2CH_3$

Solution:

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Q.13 Answer:

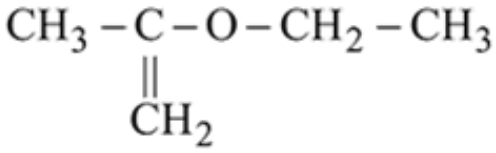
ethane

Solution:

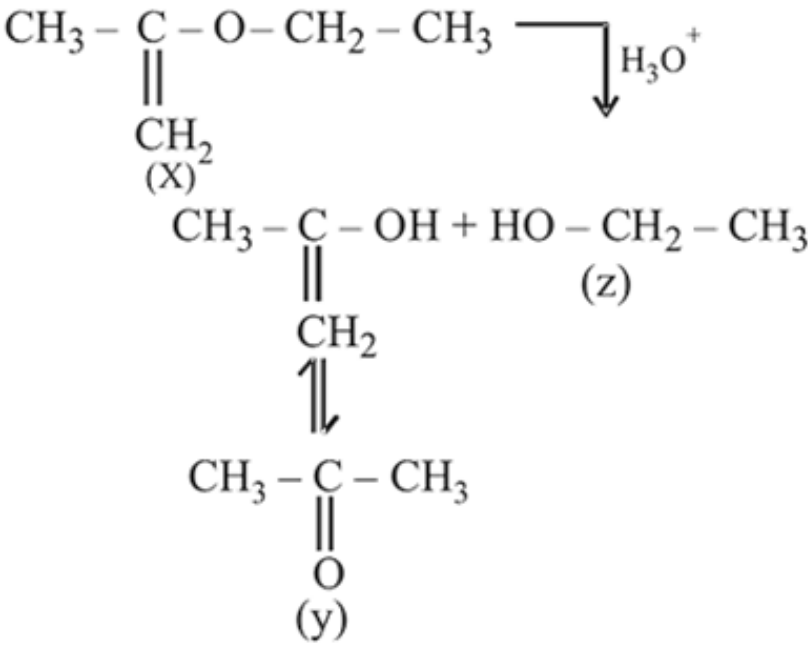
Ethers are reduced by red P and HI to alkanes through alkyl iodides



Q.14 Answer:



Solution:



Q.15 Answer:

5

Solution:

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